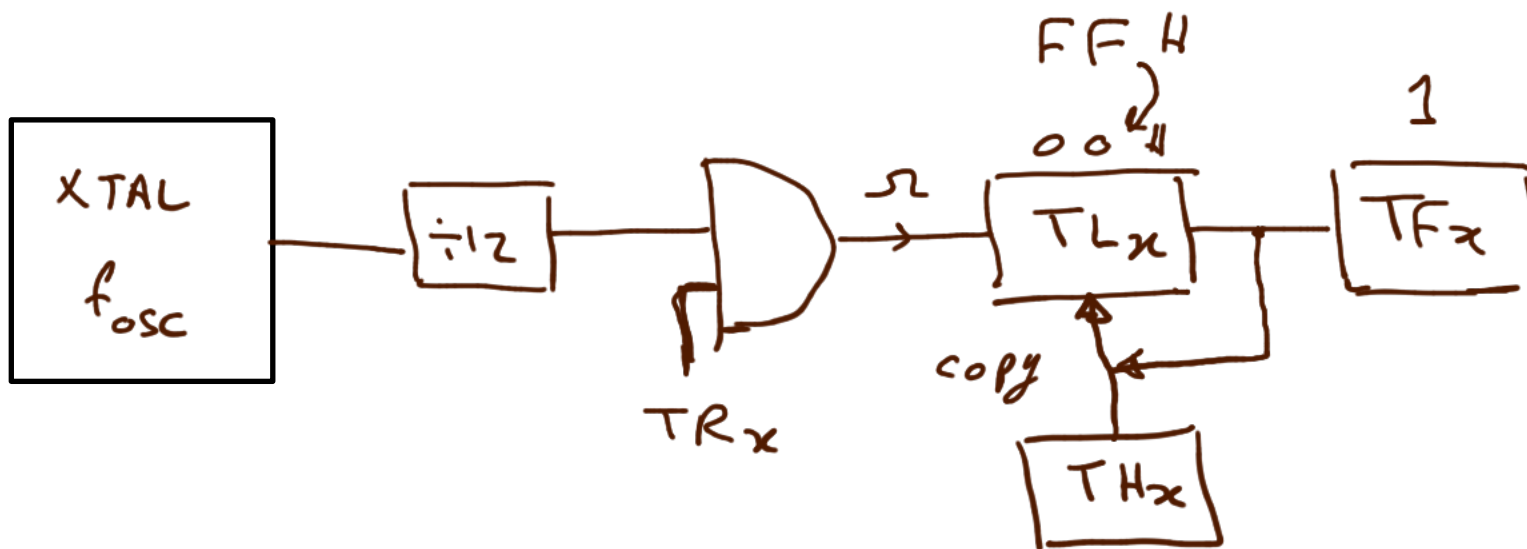


Cont. Timer

①

② Timer_x in mode 2 : is called auto reload
(use 8 bit)

range : 00H $\rightarrow$ FFH
0 $\rightarrow$ 255H



Steps in mode 2 :

① Load TMOD
MOV TMOD, # —

② load TH_x with initial value

MOV TH_x, # YYH
?

③ start timer

SETB TR_x

The timer counts until FFH then it rolls over to 00H then TF_x = 1.

④ Monitoring TF_x using

HERE: JNB TF_x, HERE

When TF_x = 1 then 8051 autoreload TL_x from TH_x.

⑤ When TF_x = 1 you have to clear TF_x

CLR TF_x

CLR TR_x optional

⑥ To repeat the process go to step ③ or ④

* calculate value to be loaded in TH_x

$$① \quad MC = \frac{12}{f_{osc}}$$

$$② \quad \frac{\text{Delay}}{MC} = \underline{\underline{Z}} \leq 256 \longrightarrow \underline{\underline{\text{mode 2}}}$$

③ options:

→ (a) `MOV THx, # -Z` ←

→ (b) $256 - Z = K$

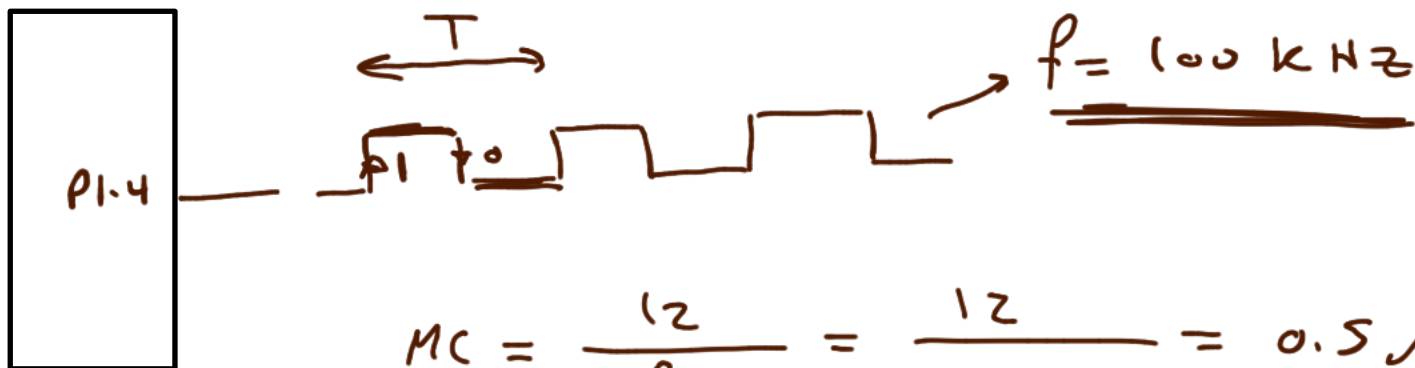
`MOV THx, # K` ←

→ (c) Convert K to Hex = YYH

`MOV THx, # YYH` ←

Write a program to create square wave of 100 kHz

on P1.4. Assume crystal frequency = 24 MHz . Use timer 1.



$$MC = \frac{12}{f_{osc}} = \frac{12}{24\text{ MHz}} = 0.5\text{ }\mu\text{s}$$

$$\frac{\text{Delay}}{MC} = \frac{?}{0.5\text{ }\mu\text{s}} = \frac{0.005\text{ ms}}{0.5\text{ }\mu\text{s}} = 10 \leq 256$$

↓
mode 2

$$T = \frac{1}{f} = \frac{1}{100\text{ kHz}} = 0.01\text{ ms}$$

$$\text{Delay} = \frac{T}{2} = \frac{0.01\text{ ms}}{2} = 0.005\text{ ms}$$

(a) $\rightarrow$ MOV TH1, # -10

(b) $\rightarrow$ 256 - 10 = 255

MOV TH1, # 255

MOV TMOD, # 0010 0000 B $\xrightarrow{20H}$

MOV TH1, # 255 $\rightarrow$ (b)

Back : SETB TRI

HERE : JNB TFI, HERE

CPL PI.4

CLR TFI

CLR TRI

SJMP Back

MOV TMOD, # 20H
MOV TH1, # 255

Back : SETB PI.4
ACALL DELAY
CLR PI.4
ACALL DELAY
SJMP Back

Back : CPL PI.4
ACALL DELAY
SJMP Back

DELAY: SETB TRI
HERE : JNB TFI, HERE
CLR TFI
CLR TRI
RET

delay

* calculate delay:

$T_{MOD} = ?$

$\Rightarrow \square$

(5)

Options:

→ (a) $MOV\ THx, \# - Z$

$$Delay = Z * MC$$

→ (b) $MOV\ THx, \# K$

$$Delay = (256 - K) * MC$$

→ (c) $MOV\ THx, \# Y Y H$

$$Delay = (FFH - YYH + 1) * MC$$

Find frequency?

Timer 1 $\rightarrow$ mode 2

(b) $\rightarrow TH1 = 5$

$$\rightarrow delay = (256 - 5) MC$$

$$= 251 \times 1.085 \mu s$$

$$= \Delta = 272.5 \mu s$$

$$T = 2 * delay = 2\Delta \rightarrow f = \frac{1}{T} = \frac{1}{2\Delta} = \square$$

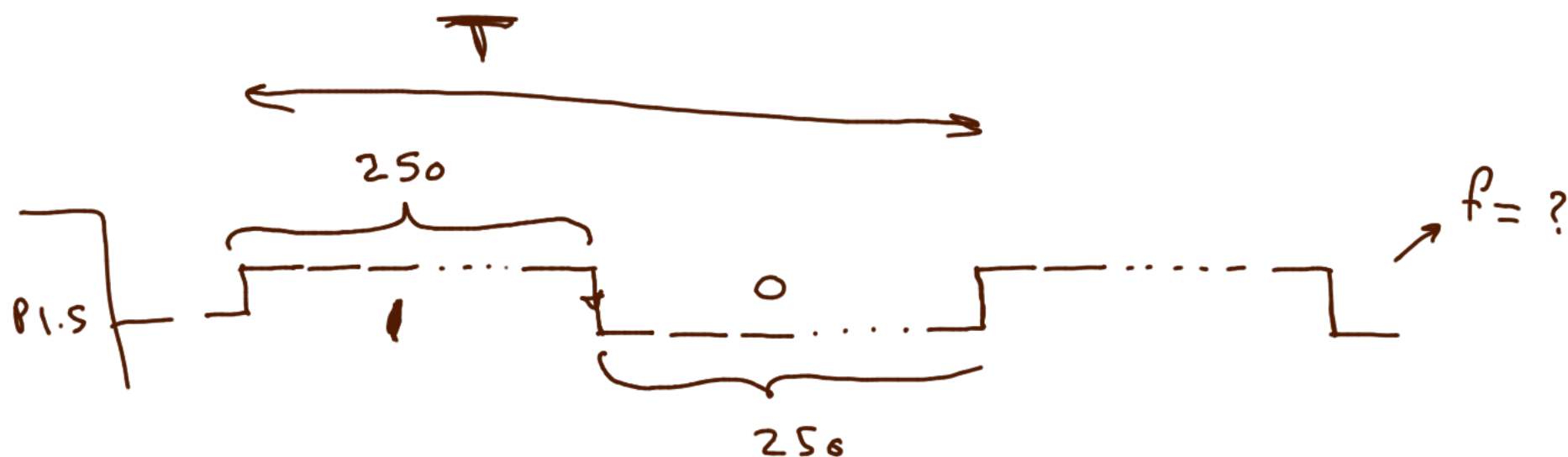
⑥ smallest freq.

min $f \longrightarrow$ max $T \longrightarrow$ max delay

$$\text{Delay} = (256 - \underline{0}) \text{ MC} \\ = 256 * \text{MC}$$

max $f \longrightarrow$ min $T \longrightarrow$ min delay

$$\text{Delay} = (256 - \underset{\substack{\downarrow \\ \text{FFH}}}{255}) \text{ MC} \\ = \text{MC}$$



$$T = 500 * \text{delay}$$

$$f = \frac{1}{T} =$$

$$\text{delay} = (256 - 0) \text{ MC} \\ = 256 * \text{MC} \\ = \Delta$$

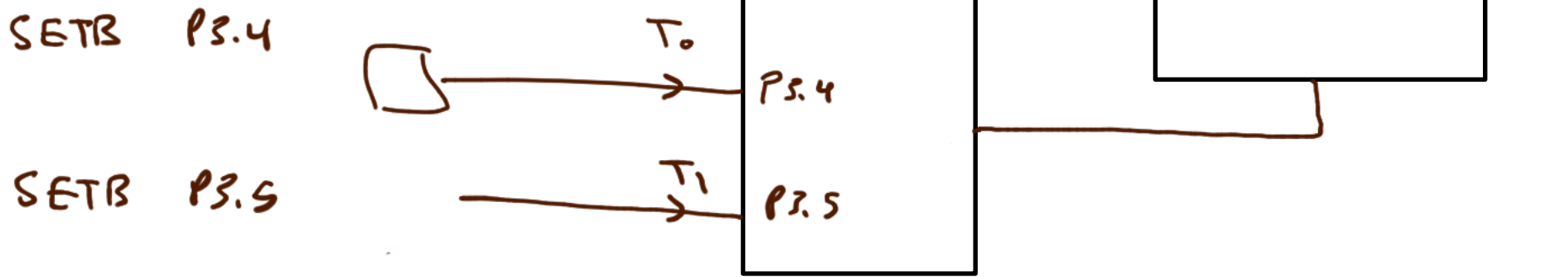
Counter

⑦

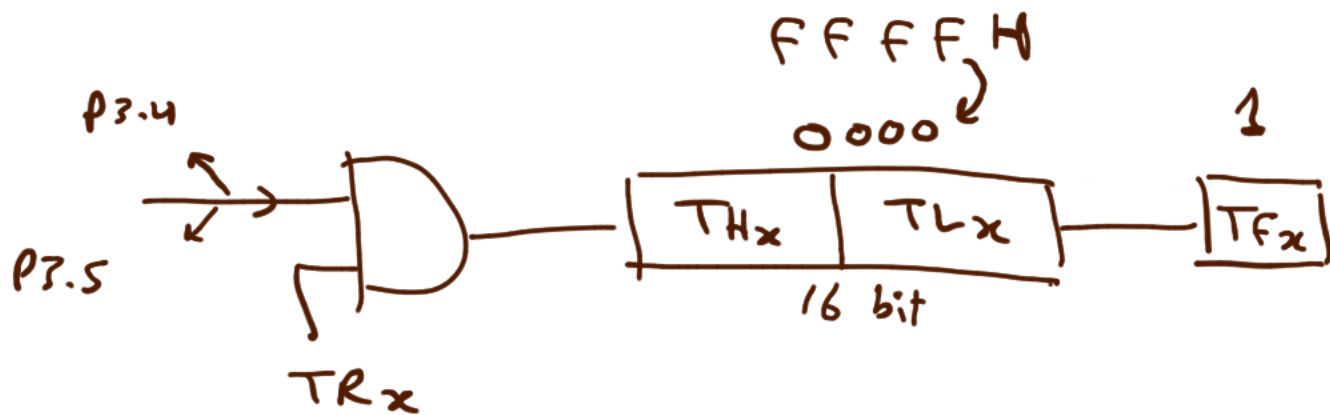
- 8051 has 2 counters

- Counter 0 (T0) = P3.4
- Counter 1 (T1) = P3.5

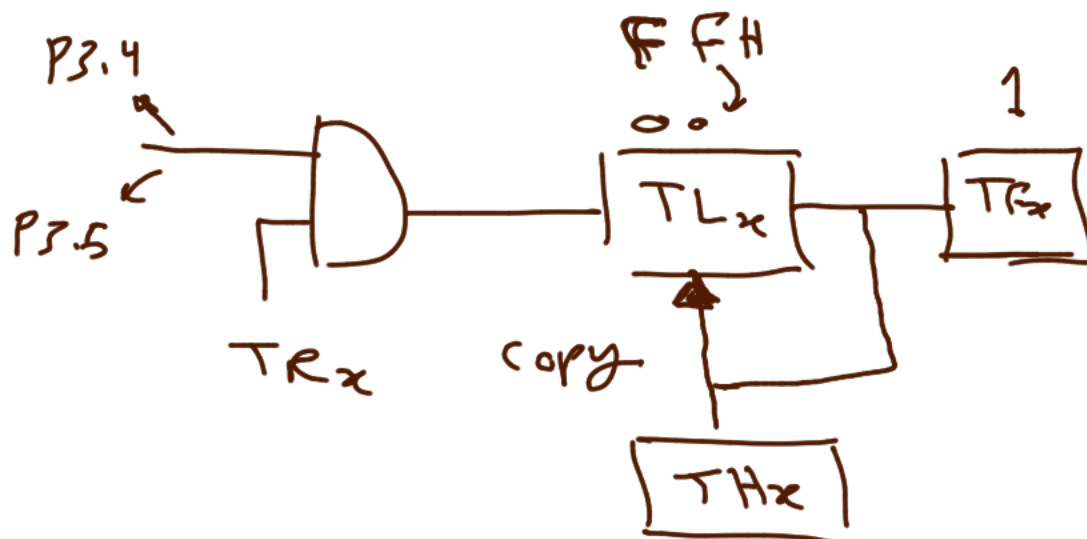
you have to make it
input pin



① Counter x in mode 1

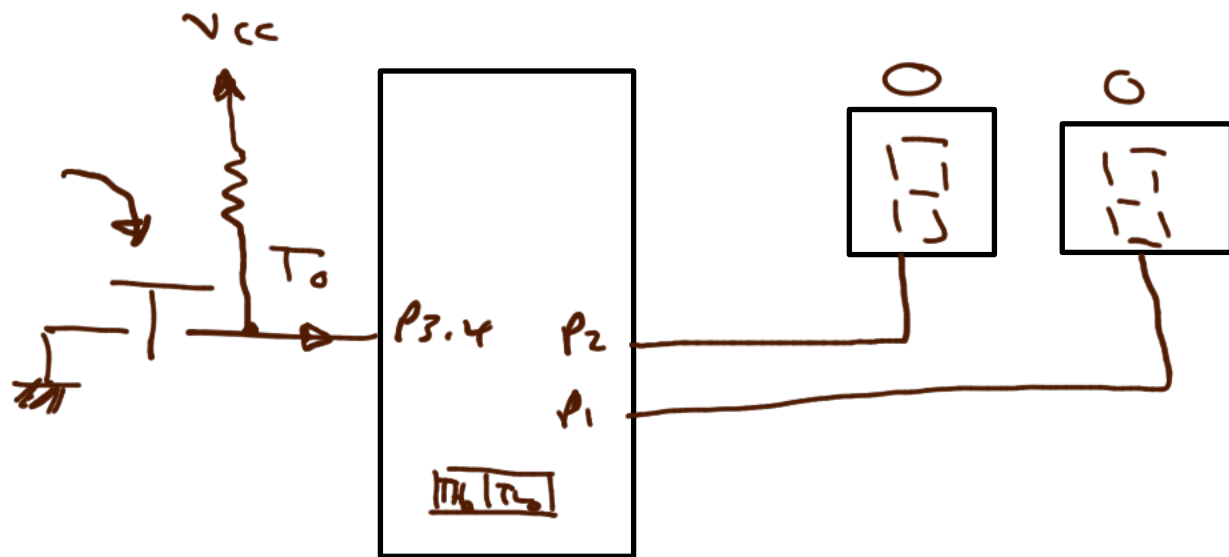


② Counter x in mode 2 (8 bit)



Write a program to count pulses fed to T₀. Display counts on P₁ (low byte) and P₂ (high byte).

②



ORG 0 → 05H

MOV TMOD, #0000 0101B

AGAIN: MOV TLO, #00H

MOV TH0, #00H

SETB P3.4

SETB TR0 ; start counter

Back: MOV P1, TLO

MOV P2, TH0

JNB TF0, Back

CLR TF0

CLR TR0

SJMP AGAIN

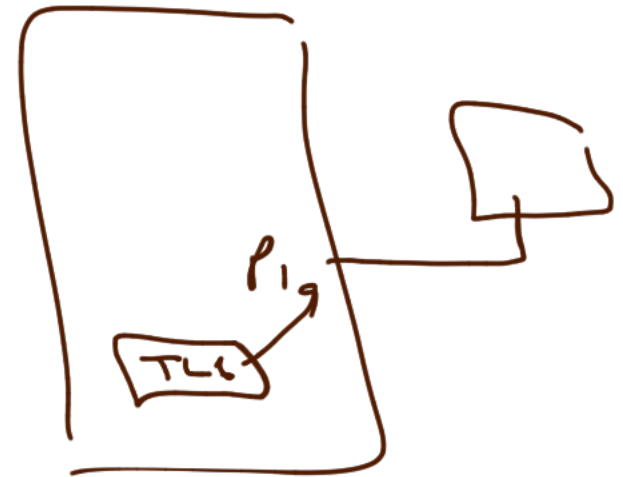
write a program to count pulses fed to T₁ (P3.5)

⑨

display counts on P1.

mode 2

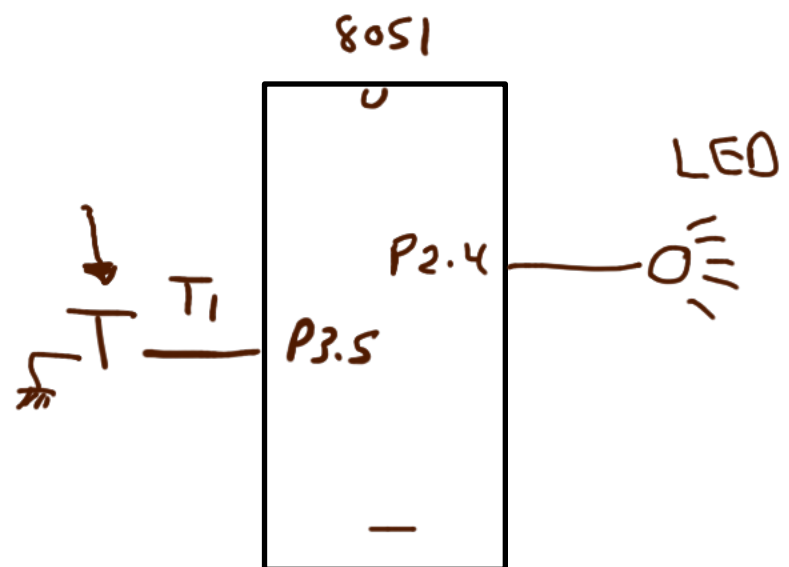
```
MOV    TMOD, #0110 0000 B  
MOV    TH1, #0  
SETB   P3.5 ; input  
SETB   TR1  
HERE : MOV    P1, TL1  
        JNB    TF1, HERE  
        CLR    TF1  
        SJMP   HERE
```



Write a program to turn on LED connected to P2.4 after 15 pulses fed to P3.5.

mode 2

counter 1



ORG 0 → 60H
MOV TMOD, # 0110 0000 B

256 - 15 = 241

MOV TH1, # -15 ←

SETB P3.5

CLR P2.4 ; LED off

SETB TR1 ; start counter

HERE : JNB TF1, HERE

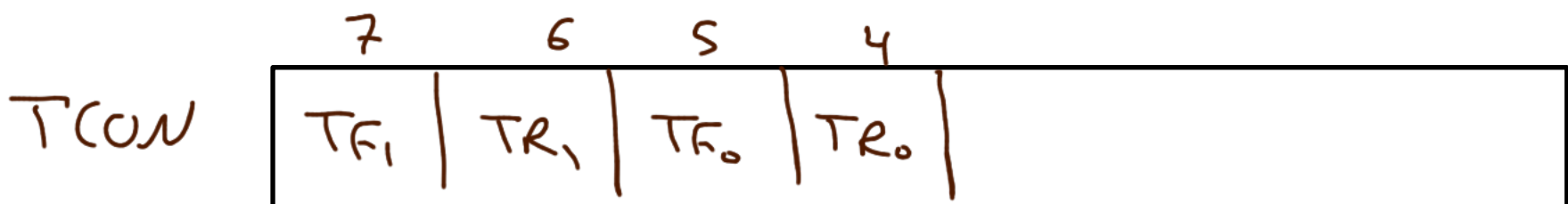
CLR TF1

CLR TR1

SETB P2.4 ; LED on

SJMP \$

TCON : Timer control register (8 bit) → bit addressable



→ SETB TR₀
→ SETB TCON.4

HERE : JNB TCON.7, HERE

(11)

- Timer
- to generate time delay
 - The source of pulses is crystal oscillator

- counter
- to count outside events
 - The source of pulses is outside device.

When $TF_x = 1$ (when the timer roll over) ?

① mode 1 : rolls over from FFFFH to 0000H

② mode 2 : " " " FFH to 00H

③ mode 0 : " " " 1FFFFH to 0000H

